

Amin Draidrya

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PROFILE

ALM Analyst with a strong **Data Science** focus at **CDG Capital**, specialized in building data pipelines (collection, cleaning, data quality control) and forecasting/time-series models applied to macro-financial indicators and **liquidity risk**. Experienced in delivering decision-support dashboards (**Streamlit/Plotly**) and early-warning scores (EWI) to support ALM monitoring and decision-making.

EDUCATION

Moroccan School of Engineering Sciences <i>Engineering Degree, Applied IT Methods for Business Management</i>	Casablanca, Morocco 2022 – 2025
Preparatory Classes for Engineering Grandes Écoles <i>MPSI/MP Track</i>	Rabat, Morocco 2019 – 2022

PROFESSIONAL EXPERIENCE

ALM Analyst <i>CDG Capital</i>	September 2025 – Present Casablanca, Morocco
- Designed and developed an <i>Early Warning Indicators (EWI)</i> dashboard in <i>Python (Streamlit/Plotly)</i> automating ingestion (API/scraping), cleaning, historization, and <i>scoring</i> of macro-financial indicators (<i>MONIA, BDT, reserve requirements, MASI, FX</i>) to generate <i>alerts</i> and a market-bias summary for ALM. - Designed and calibrated <i>DDA (demand deposits)</i> models: <i>stable/unstable</i> segmentation, <i>run-off</i> curves (Selvaggio/Markov), sensitivity analysis, and backtesting to strengthen <i>liquidity risk</i> management (ALM/LCR).	
Final-Year Internship <i>Caisse de Dépôt et de Gestion (CDG)</i>	March 2025 – June 2025 Rabat, Morocco
- Developed forecasting models for institutional clients' <i>balances</i> (<i>ARIMA/LSTM/GRU</i> approaches) to anticipate liquidity variations and identify high-risk profiles for large withdrawals (instability, concentration, atypical behaviors). - Time-series feature engineering (lags, monthly aggregations, returns) and evaluation via backtesting (MAE/RMSE), with residual analysis to improve forecast robustness.	
End-of-Year Internship <i>V&G Capital Solutions</i>	August 2024 – September 2024 Casablanca, Morocco
- Built an interactive macro-indicators dashboard (Python/Plotly) that reduced analysis time by <i>20%</i>, featuring dynamic visualizations and automatic comment generation for decision support. - Implemented a <i>reward/penalty</i> mechanism (reinforcement-learning-inspired logic) to adjust rules for predicting market bias and central bank decisions based on historical performance.	
End-of-Year Internship <i>Huawei Technologies Co., Ltd</i>	July 2023 – August 2023 Casablanca, Morocco
Automated a customer communication and support workflow, improving team productivity by <i>15%</i>, and participated in a <i>Cloud Computing</i> training program.	

PROJECTS

Real-Time Database Replication | *SQL Server, Kafka, CDC, ETL*

- Designed and implemented a **real-time** replication pipeline from a production SQL Server database to a backup database to ensure availability and business continuity.
- Captured changes (CDC) and streamed events through **Kafka** (topics), with consumption on the target side to apply updates in an ordered and reliable manner.
- Handled robustness aspects: incident recovery, idempotence, data consistency, and flow monitoring (latency/lag).

Volatility Modeling with GARCH (S&P 500) | *Python, yfinance, arch, Statsmodels, Matplotlib*

- Analyzed and forecasted S&P 500 volatility using GARCH-family models for market risk assessment.
- Collected historical data (2000–2025) via the Yahoo Finance API, computed daily returns, and prepared series for modeling.

- Implemented and compared **GARCH(1,1)** and **GJR-GARCH** models to capture asymmetry (leverage effect) in volatility.
- Visualization and interpretation: estimated conditional volatility, diagnostics, and summary charts with Matplotlib.

Portfolio Optimization Web App (MPT) | *Python, Streamlit, PyPortfolioOpt, Yahoo Finance, Plotly*

- Developed an interactive web tool for portfolio optimization based on Modern Portfolio Theory (mean-variance) to recommend optimal allocations given risk/return preferences.
- Automatically retrieved 5 years of price history (user-input tickers) via the Yahoo Finance API, and computed expected returns and the covariance matrix.
- Optimization with PyPortfolioOpt: built the efficient frontier, selected the optimal portfolio, and fell back to Sharpe maximization when the target return was unattainable.
- Ran a Monte Carlo simulation (5,000 random portfolios) and visualized the efficient frontier with metrics (return, volatility, Sharpe) and optimal weights.

Gold Price Modeling & Forecasting | *Python, Statsmodels, Scikit-learn, TensorFlow, ARIMA, LSTM*

- Built a gold price forecasting framework using **statistical** and **deep learning** approaches for time series.
- Data preparation: cleaning, transformation, lagged-feature creation, and time-based train/validation splitting to avoid data leakage.
- Implemented and compared **ARIMA** (baseline) and **LSTM** (nonlinearities), evaluated using (MAE/RMSE), and performed residual analysis.
- Produced visualizations and performance reporting (forecast vs actual) to support interpretation and iterative model improvement.

TECHNICAL SKILLS

Programming: Python, SQL, C++
Data Science & ML: Feature Engineering, Model Selection & Cross-Validation, Deep Learning, Time Series & Forecasting, Reinforcement Learning
Risk/Finance Modeling: Risk Modeling (Market/Liquidity), Stress Testing, ALM/LCR/NSFR, Pricing (basics), Portfolio Optimization
Data Engineering: ETL/ELT, Data Warehousing, Pipelines (Airflow/Prefect), Data Modeling, Orchestration
Databases: PostgreSQL, SQL Server, SQL Tuning
BI & Visualization: Power BI, Plotly, Matplotlib
APIs & Web Data: REST APIs, Web scraping (Requests/BS4/Selenium), FRED, Yahoo Finance
Tools: Git, Linux, Jupyter, VS Code, Excel (VBA - basics)

LANGUAGES AND STRENGTHS

Languages	Strengths
• Arabic: Native • French: Advanced • English: Advanced	• Problem solving • Adaptability • Analytical mindset